lipids, that forms our cell membrane and the glycogen the energy reserve stored in our liver are all organic compounds. Except few inorganic compounds like salt, water etc... all others such as food, medicine, clothing, cosmetics, fuel etc... are compounds of carbon. All the essential biochemical reactions are also organic in nature leading to the fomation of various essential bioorganic molecules such as lipoproteins, phospholipids, glycolipids etc...

Synthesis of acetic acid by kolbe and methane by Berthlot, confirmed that organic compounds can be synthesized in laboratory. Since then, millions of organic compounds were synthesised and characterised. The field of organic chemistry is very vast and its principles find applications in many industries including food, textile, petrochemical, pharmaceutical, dye, polymers, fertilizer, cosmetics etc... Discussing the importance of organic chemistry is just like describing a drop of water in a mighty ocean.

The knowledge of chemical bonding and molecular structure will help in understanding the properties of organic compounds. We know that, the carbon has four valance electrons and its ground state electronic configuration is $1s^2\ 2s^2\ 2p^2$. An atom can attain noble gas electronic configuration either by transferring or sharing of electrons. It is not possible for the carbon to form either C^{4+} or C^{4-} ions to attain the nearest noble gas configuration, as it requires large amount of energy. This implies that carbon cannot form ionic bond. Almost in all compounds of carbon, it forms four covalent bonds.

The formation of four covalent bonds can be explained as below. During bond formation, one of the electrons from 2s orbital is promoted to $2p_z$ orbital. The formation of four sigma bonds by carbon can be explained on the basis of sp^3 hybridisation of carbon. Carbon forms multiple (double and triple) bonds in certain compounds. These can be explained by sp^2 and sp hybridisation of carbon. The carbon forms relatively short bonds which enable the lateral overlap of unhybridised 2p orbitals of sp^2 and sp hybridised carbon to form one and two pi bonds respectively.

Molecular stucture can be derived from the type of hybridisation. An sp^3 hybridised carbon will have a tetrahedral geometry, a sp^2 hybridised carbon will have trigonal planar geometry. and sp hybridised carbon will have a linear geometry.

Characteristics of organic compounds:

All organic compounds have the following characteristic properties.

1. They are covalent compounds of carbon and generally insoluble in water and readily soluble in organic solvent such as benzene, toluene, ether, chloroform etc...

2. Many of the organic compounds are inflammable (except CCl_4).They possess low boiling and melting points due to their covalent nature

3. Organic compounds are characterised by functional groups. A functional group is an atom or a specific combination of bonded atoms that

11th Chemistry Chapter-11 Vol-2 EM.indd 111 09-01-2023 14:42:55